Adnan Shahriar, PhD

Phone: 210-803-5230 | Email: shahriar.adnan@yahoo.com
LinkedIn: https://www.linkedin.com/in/adnan-shahriar/
GitHub: https://github.com/AdnanShahriar64
Google Scholar: https://scholar.google.com/citations?user=jF2HLEcAAAAJ

PROFESSIONAL SUMMARY

Ph.D. mechanical engineer combining structural and fluid simulation, nonlinear material modeling, and experimental validation. Developed custom three-dimensional FEM/XFEM and coupled multiphysics solvers, calibrated structural models against measured response, and applied explicit dynamic impact analysis to plastic-component cracking. In product development, used CFD/FEA, parametric automation, and prototype testing to evaluate design alternatives and improve structural and airflow performance.

TECHNICAL SKILLS

Structural Simulation: FEM/XFEM, Nonlinear material modeling, Explicit dynamic FEA, Modal analysis, J-integral assessment

Fluid and Multiphysics: CFD, Navier-Stokes equations, Transient heat transfer, Fluid-structure interaction, Coupled thermal-fluid models

Numerical Software: C/C++, Python, CadQuery, Custom solver development, Reduced-order modeling

Experimental Validation: Prototype testing, Impact-hammer testing, Structural model calibration

PROFESSIONAL EXPERIENCE

Project Engineer — Rheem Manufacturing (Friedrich Air Conditioning) | San Antonio, TX *Mar 2025–Present*

- **Explicit Dynamic Impact and Fracture Assessment**
  - Performed explicit dynamic FEA impact analysis of plastic components to evaluate impact-induced stresses and potential cracking; calculated the J-integral to assess the driving force for crack propagation.
- **Multi-Material Product FEA Analysis**
  - Performed finite element analysis (FEA) to support product designs using ABS and ASA plastics, aluminum and steel sheet metal, and expanded polypropylene (EPP) at different densities.
- **Structural FEA and Assembly Optimization of Modular AHU**
  - Performed comprehensive finite element analysis (FEA) of the nine-segment modular AHU assembly to evaluate structural integrity, press-fit connections, and load-bearing capacity. Analyzed deformation under fan and motor loads to maintain critical clearances between the rotating fan lip and stationary bellmouth, preventing mechanical interference and supporting reliable prototype operation.
- **Motor-Mount Structural Optimization**
  - Redesigned a motor-mount structure using FEA, reducing material usage by more than 50% while maintaining stiffness and vibration requirements. Simplified injection-molded components to reduce secondary manufacturing operations and assembly complexity.
- **Fan-Induced Vibration and Modal Analysis**
  - Performed FEA-based modal analysis to identify natural frequencies of product components and assess resonance risk by comparing them with fan rotational frequency (RPM/60) and blade-passing frequency.
- **Outdoor Fan-Shroud Optimization**
  - Used CFD to optimize shroud geometry and improve airflow distribution across the condenser coil. Fabricated and tested a prototype, demonstrating more than 80% improvement in airflow efficiency (CFM/W) relative to the previous design.
- **Parametric Fan-Shroud Generation**
  - Developed a Python/CadQuery-based tool that converts fan diameter, coil geometry, component locations, and design clearances into 3D shroud geometry and STEP files, reducing repetitive CAD modeling and accelerating design iteration across product configurations.
- **Refrigerant-Tube Routing Optimization**
  - Developing an FEA-based tool to evaluate natural frequencies and optimize compressor-to-outdoor-coil tubing routes to reduce vibration susceptibility. Incorporating experimental modal testing and model updating to calibrate FEA predictions against measured vibration characteristics.

Senior Simulation Integration Engineer — L&T Technology Services | Plano, TX *Jan 2024–Mar 2025*

- **Modular GUI for Aerodynamic Data Analysis**
  - Developed a modular GUI for aerodynamic data analysis with well-defined input/output interfaces, improving maintainability and reducing computational and memory overhead.
- **Real-Time C/C++ HVAC Plant Model for SiL and HiL**
  - Developed a real-time C/C++ executable HVAC plant model for Software-in-the-Loop (SiL) and Hardware-in-the-Loop (HiL) applications, combining finite element solution of transient heat transfer and velocity-pressure-coupled Navier-Stokes equations with lumped-capacitance heater/cooler models, convective thermal-fluid coupling, and a proportional-derivative (PD) HVAC controller model.
- **Jenkins CI/CD and Automated Debugging**
  - Developed Jenkins CI/CD pipelines with Bash and PowerShell scripting and automated screen-capture capability, streamlining debugging and test workflows for construction and forestry vehicle software.

Graduate Researcher — Resilient Extraterrestrial Habitats Institute | San Antonio, TX *Dec 2019–Jan 2024*

- **XFEM, Nonlinear Materials, and Damage Modeling**
  - Developed XFEM methods to represent stress fields around three-dimensional voids and implemented nonlinear material behavior incorporating plasticity, strain-rate effects, thermal expansion, and von Mises and Drucker-Prager yield criteria.
- **Custom 3D Multiphysics Finite Element Solvers**
  - Developed custom three-dimensional finite element solvers from first principles for structural mechanics, transient heat transfer, and fluid dynamics, including coupled structural-thermal habitat models and interior thermal-fluid simulations.
- **Experimental Structural Digital Twin and Model Updating**
  - Created a structural digital twin of an experimental habitat by acquiring impact-hammer and accelerometer data, identifying natural frequencies and mode shapes, using Crypton optical geometry measurements, and calibrating a parameterized beam-element model to match identified modal properties and measured acceleration histories.
- **Reduced-Order Modeling and High-Order Elements**
  - Developed modal-coordinate reduced-order models and applied Guyan static condensation; implemented very high-order finite element formulations for acoustic wave propagation associated with micrometeoroid impacts.
- **Eulerian CFD and Hurricane Wave Simulation**
  - Developed Eulerian-based CFD models to simulate wave generation and fluid motion under extreme hurricane conditions, enabling numerical investigation of hydrodynamic loads on coastal bridge structures.
- **Coupled Eulerian-Lagrangian FSI Modeling**
  - Developed coupled Eulerian-Lagrangian (CEL) simulation models to investigate fluid-structure interaction between hurricane-driven waves and coastal bridge structures, integrating Eulerian fluid modeling with Lagrangian structural mechanics.
- **NASA Modular-Coupled Virtual Testbed (MCVT)**
  - Developed and integrated structural, thermal, and fluid plant models for NASA's Modular-Coupled Virtual Testbed (MCVT), a system-of-systems simulation environment for virtual testing of resilient extraterrestrial habitats.
- **Efficient Scientific Software and Numerical Algorithms**
  - Developed computational tools emphasizing reduced arithmetic and memory complexity to support novel numerical formulations and simulation workflows; contributed to a publication on algorithm development.

SELECTED TECHNICAL DEMONSTRATIONS

Large-deformation beam simulator: https://www.linkedin.com/feed/update/urn:li:activity:7429372567318331392/

CFD and bridge structural analysis: https://youtu.be/DTQD6CamibM

MCVT structural-mechanical model: https://youtu.be/TAA66P_UXIM

EDUCATION

Ph.D., Mechanical Engineering — University of Texas at San Antonio, 2023 | GPA: 3.94

B.S., Aeronautical Engineering (Aerospace Major) — Military Institute of Science and Technology, 2015